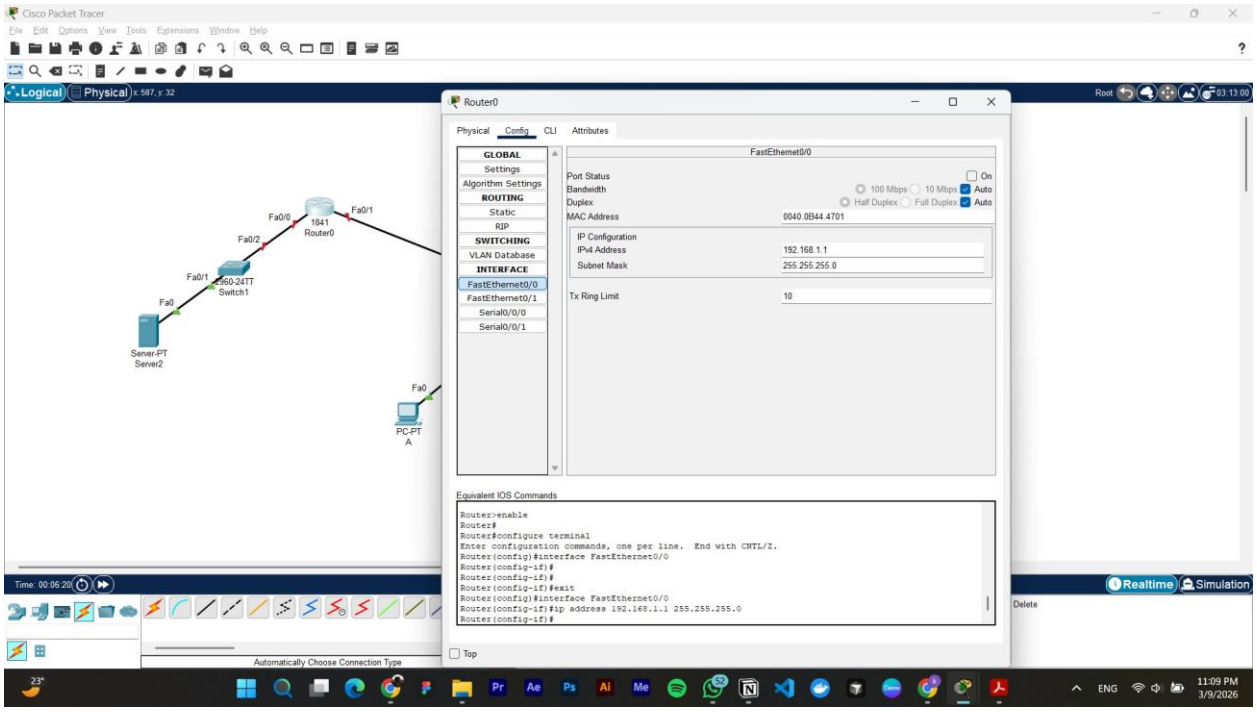
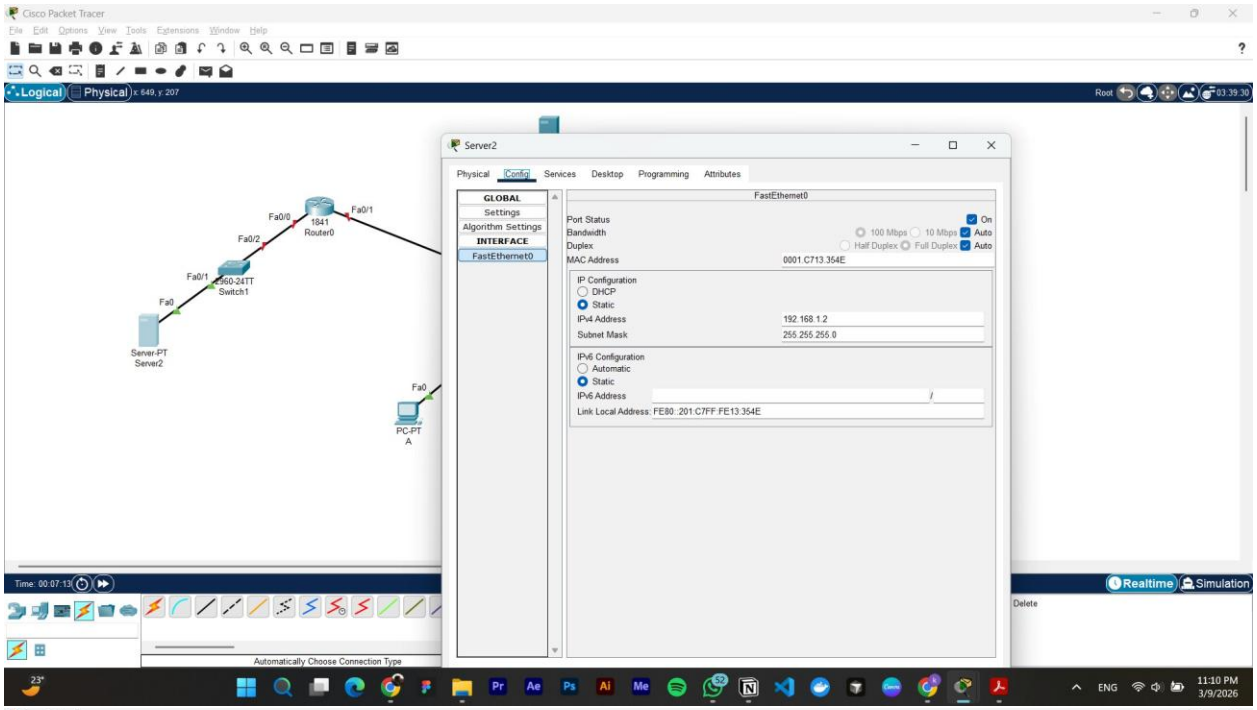
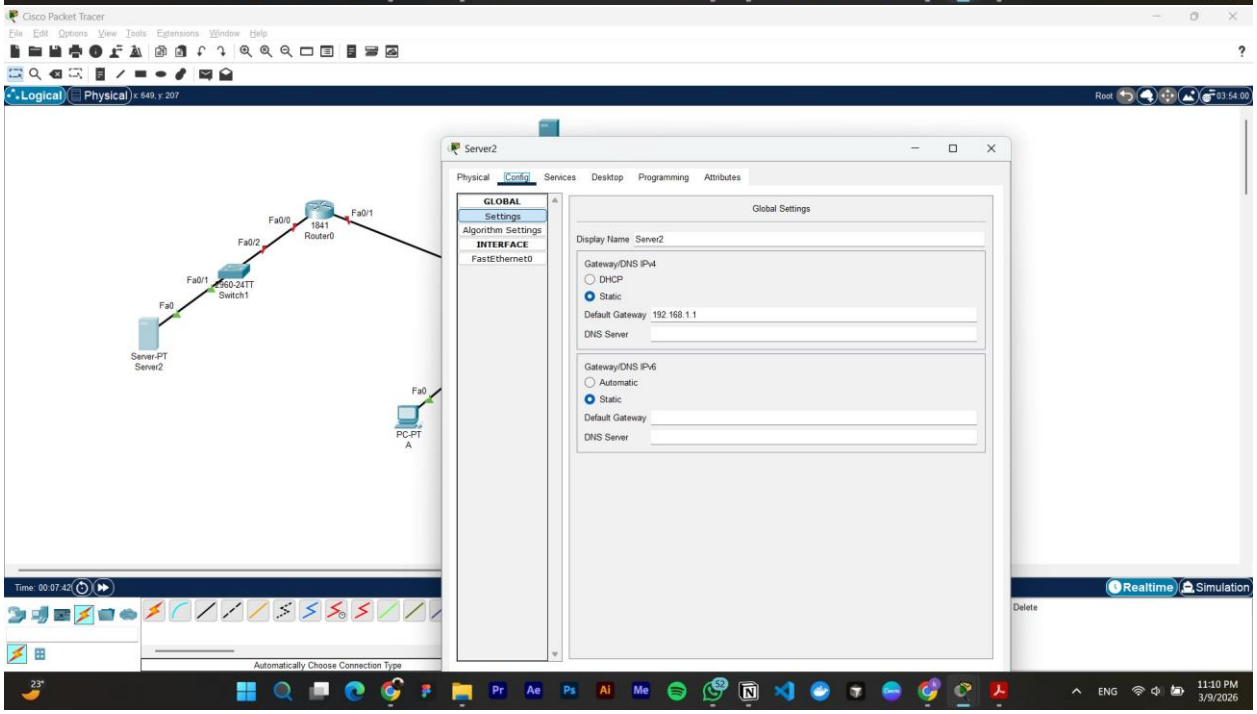
CN Lab - Lab 06 Tasks

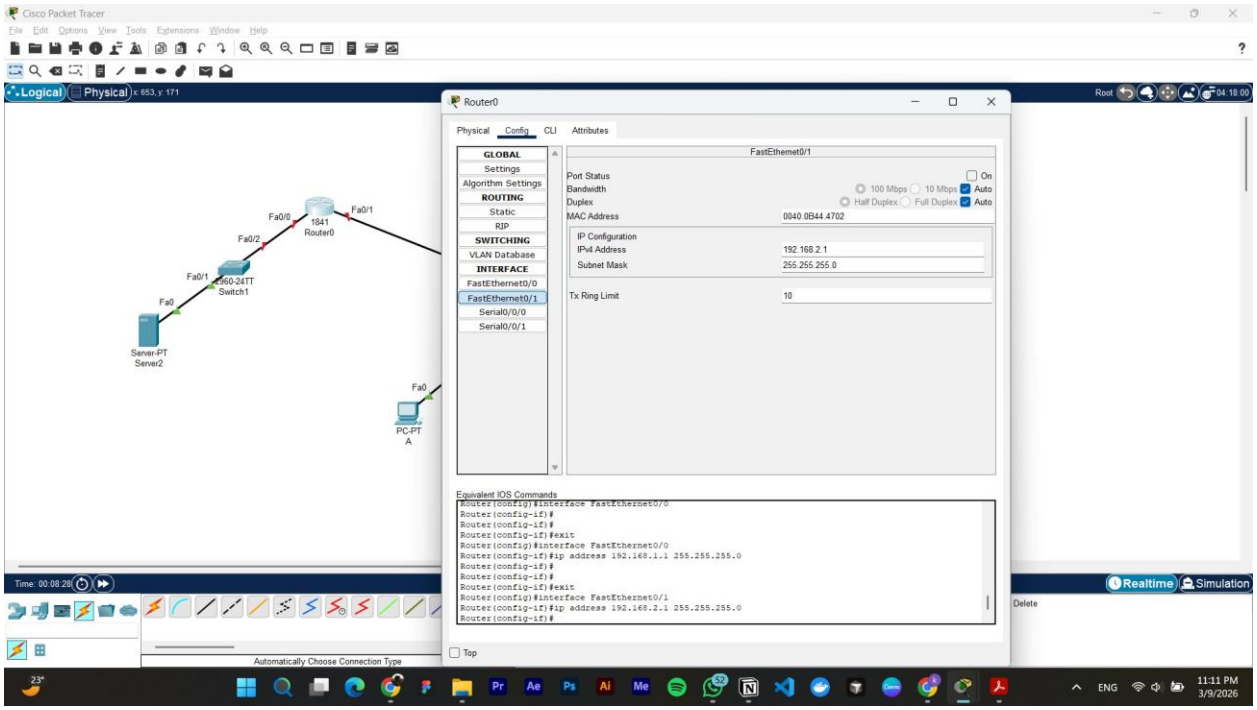
Syed Ukkashah Ahmed Shah
BAI-6A
23K-0055

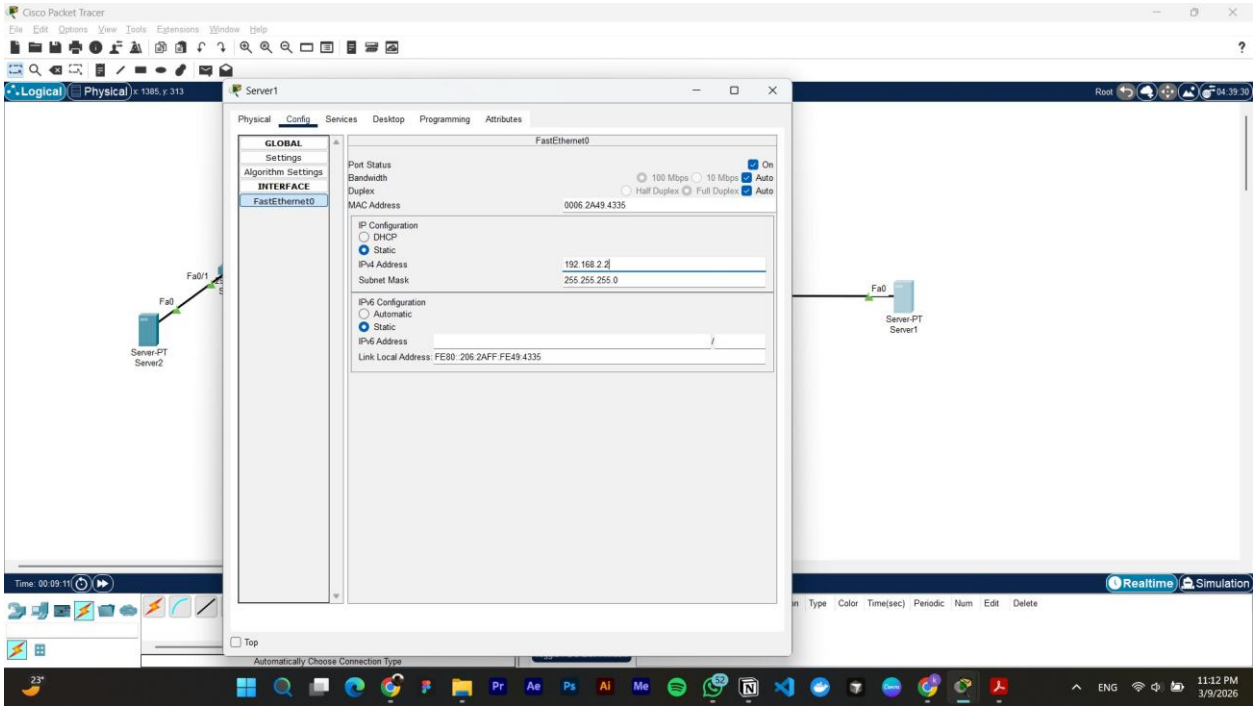
Task 1:

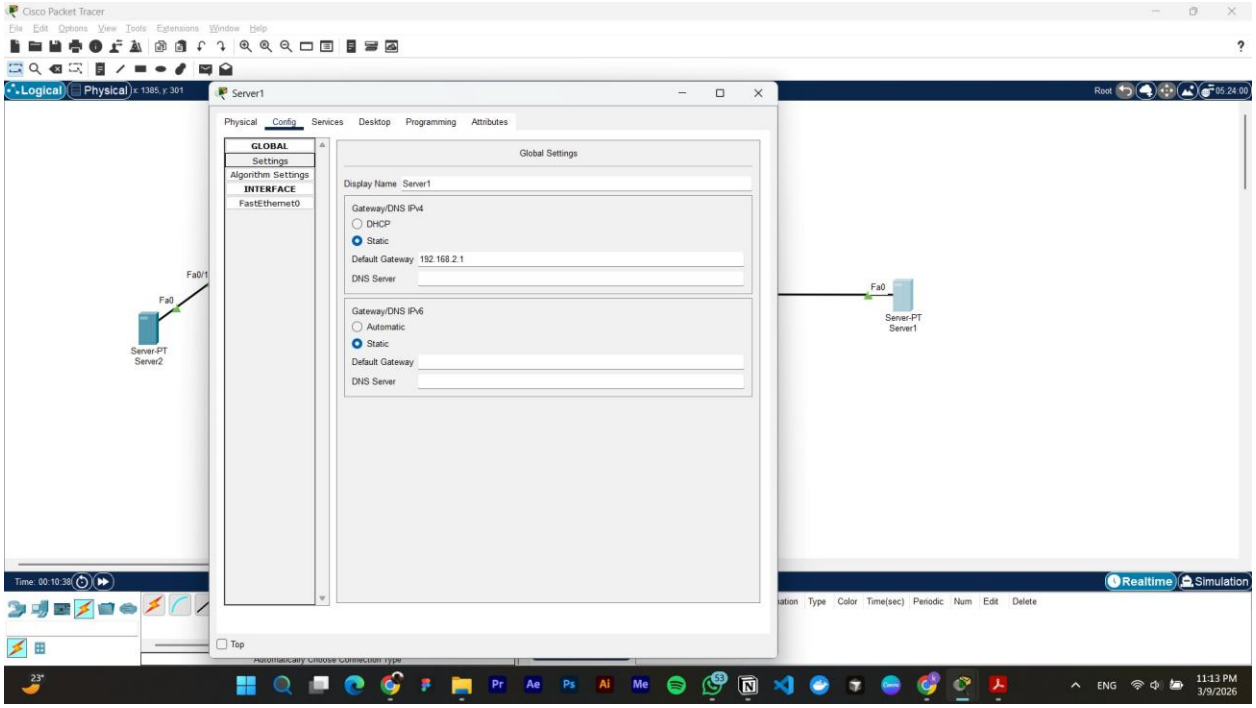


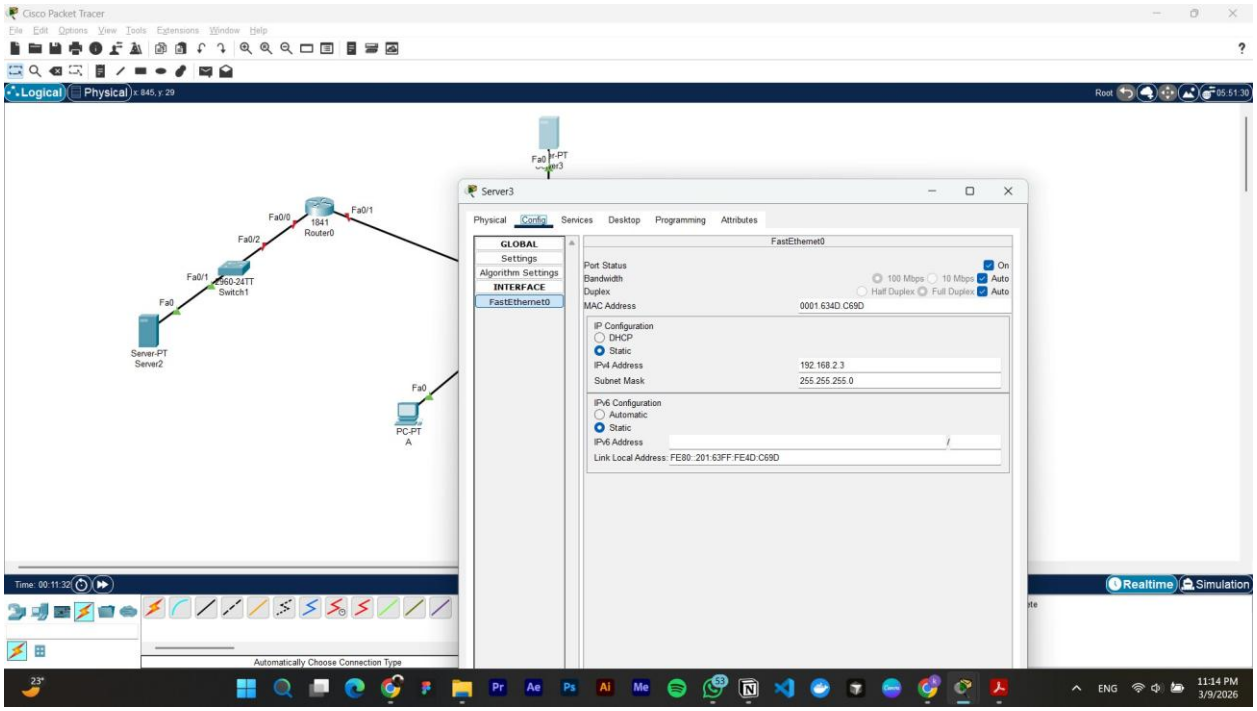

Cisco Packet Tracer
Logical
Physical
Server2
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.C713.354E
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address: FE80::201:C7FF:FE13:354E
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0
PC-PT
A
Realtime
Simulation
Delete
Automatically Choose Connection Type

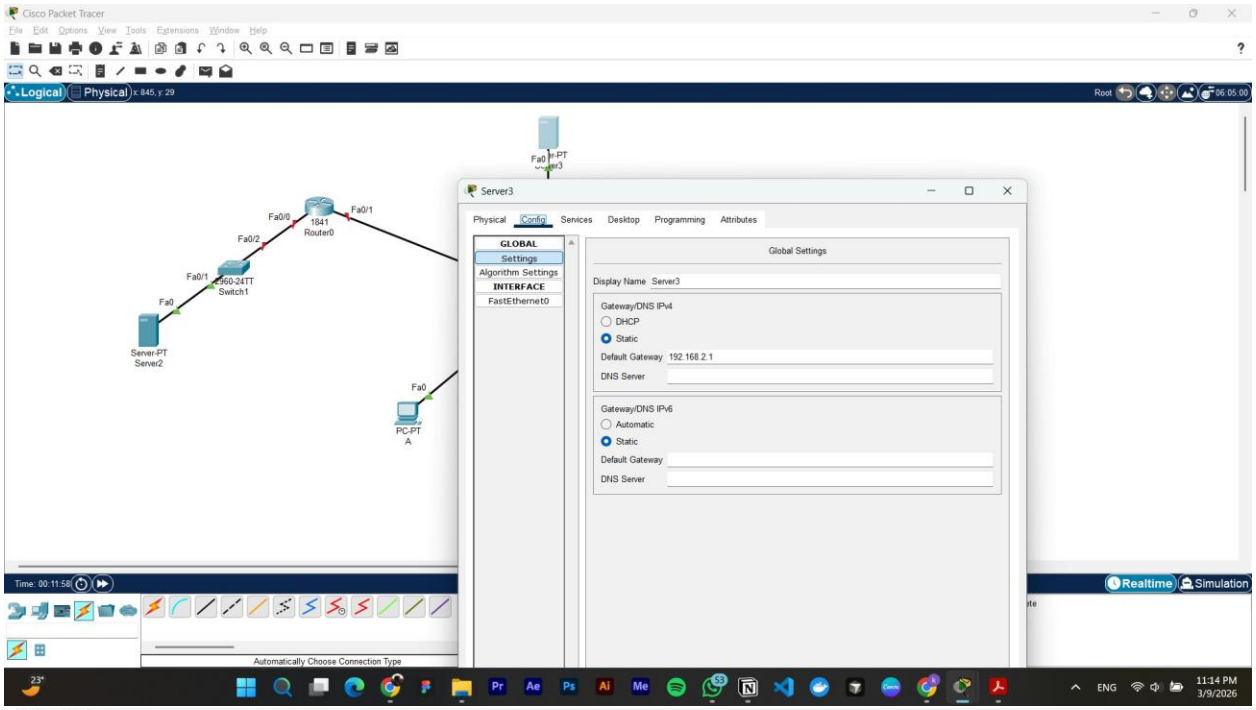

Cisco Packet Tracer
Logical
Physical
Server2
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Global Settings
Display Name
Server2
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.1.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0
PC-PT
A
Realtime
Simulation
Delete
Automatically Choose Connection Type

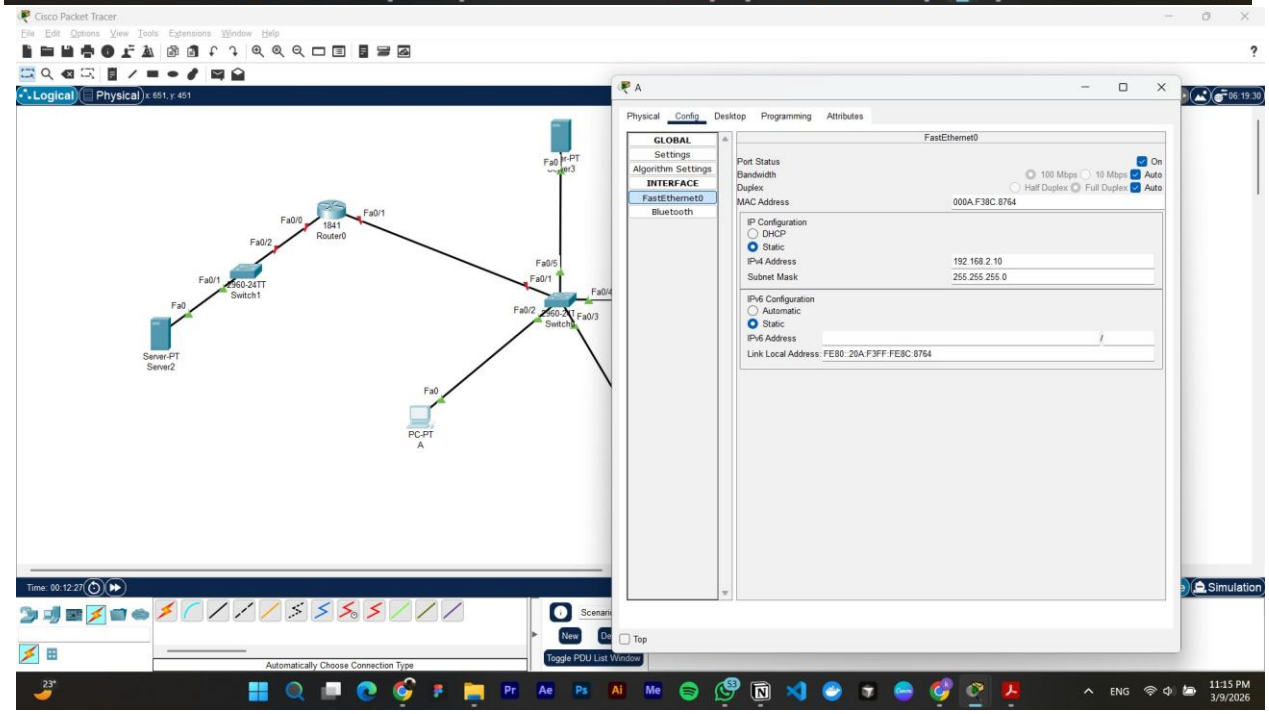

Cisco Packet Tracer
Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/0/0
Serial0/0/1
FastEthernet0/1
Port Status
Bandwidth
Duplex
MAC Address
0040.0B44.4702
IP Configuration
IPv4 Address
192.168.2.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#
Server-PT
Server2
Switch1
Router0
PC-PT
A

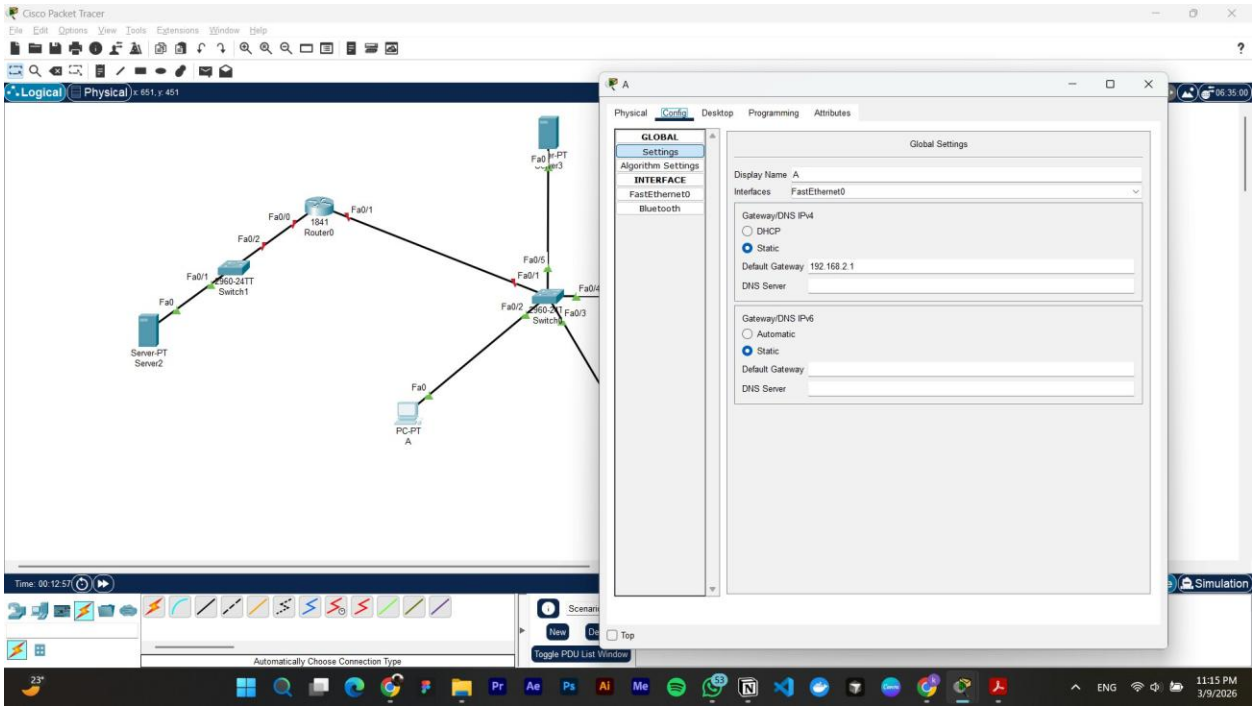

Cisco Packet Tracer
Server1
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address
0006.2A49.4335
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.2
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address: FE80::206:2AFF:FE49:4335
Server-PT
Server2
Server-PT
Server1

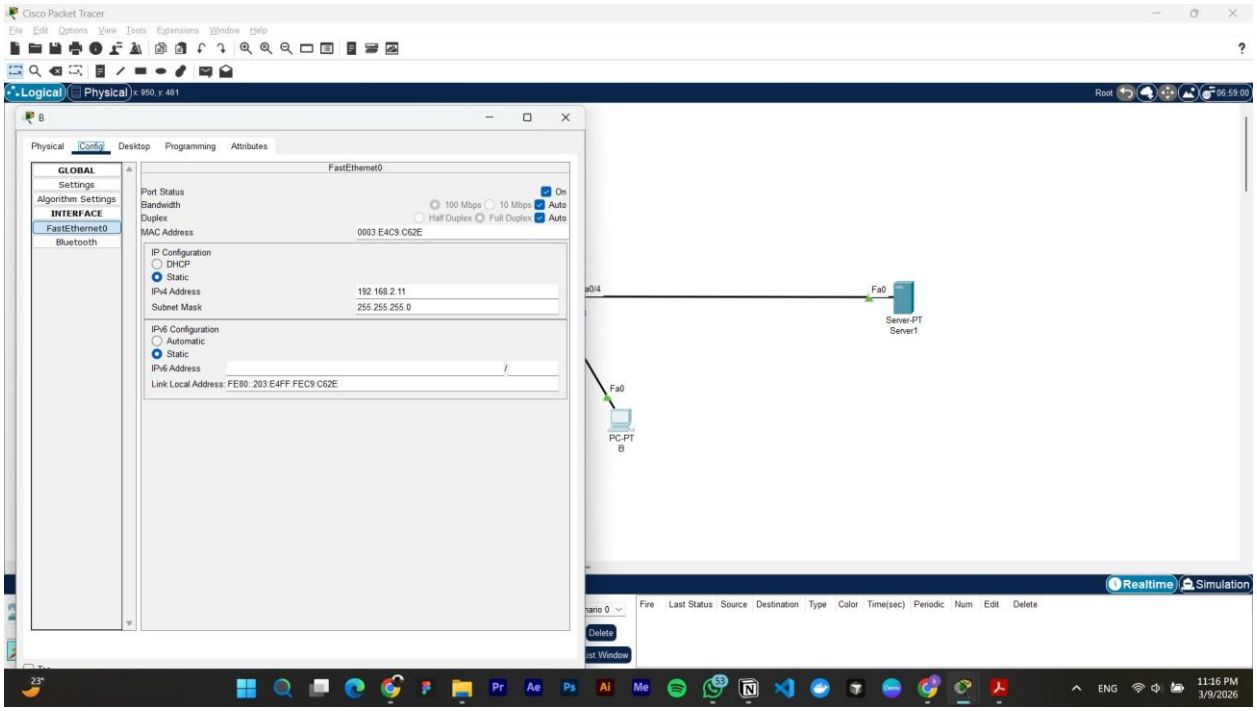

Cisco Packet Tracer
Server1
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Global Settings
Display Name Server1
Gateway/DNS IPv4
DHCP
Static
Default Gateway 192.168.2.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top
Fa0/1
Fa0
Server-PT
Server2
Fa0
Server-PT
Server1
Realtime
Simulation

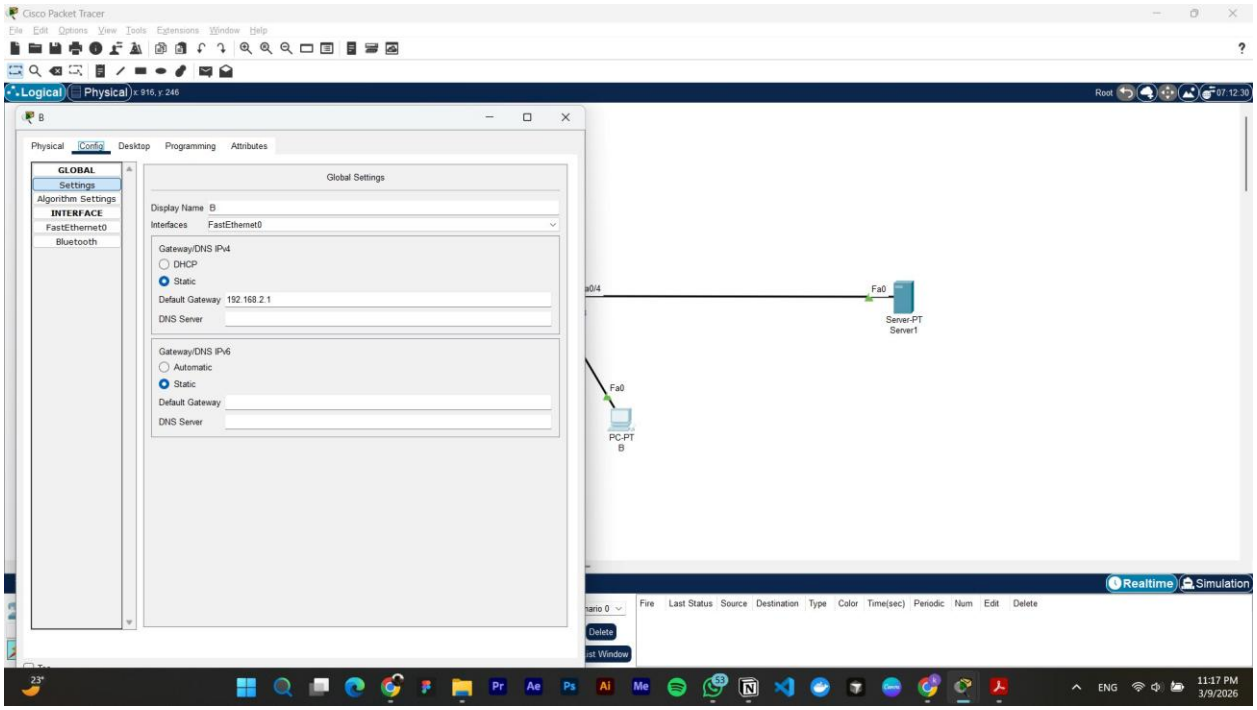

Cisco Packet Tracer
Server3
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.634D.C69D
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.3
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address: FE80::201:63FF:FE4D:C69D
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0
PC-PT
A
Realtime
Simulation
Automatically Choose Connection Type

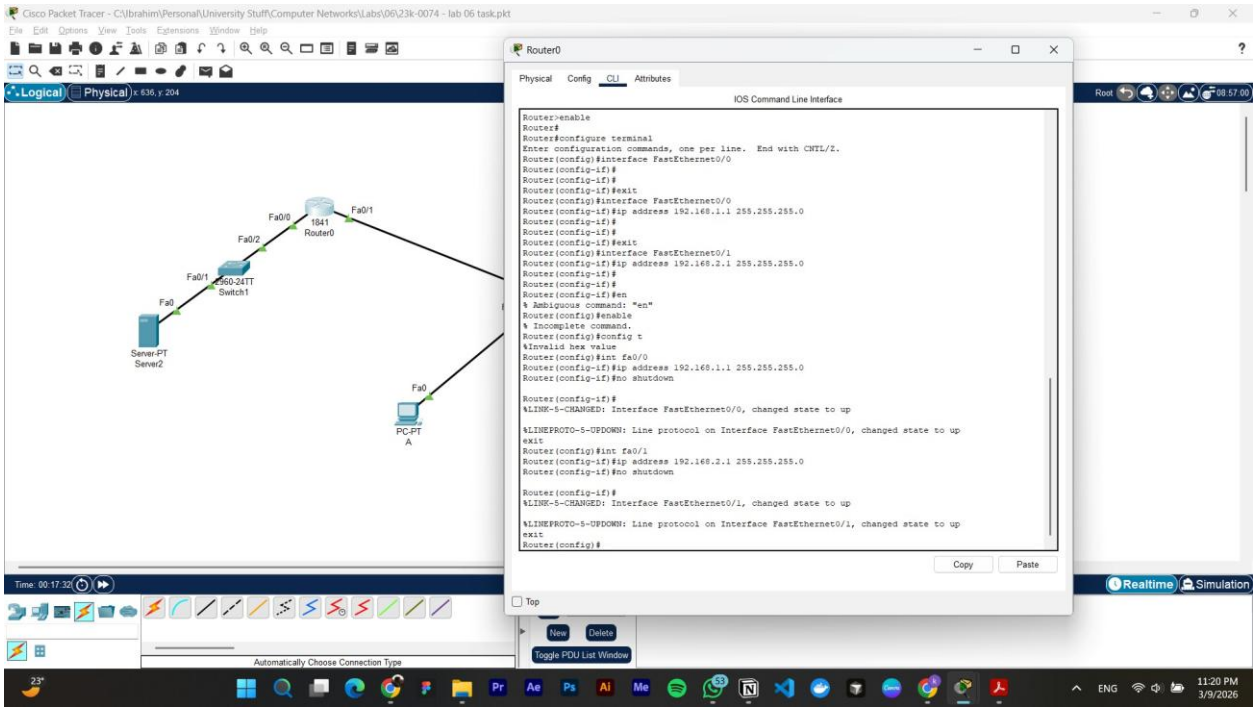

Cisco Packet Tracer
Logical
Physical
Server3
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Global Settings
Display Name Server3
Gateway/DNS IPv4
DHCP
Static
Default Gateway 192.168.2.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
1841 Router0
2960-24TT Switch1
Server-PT Server2
PC-PT A
Realtime
Simulation
Automatically Choose Connection Type

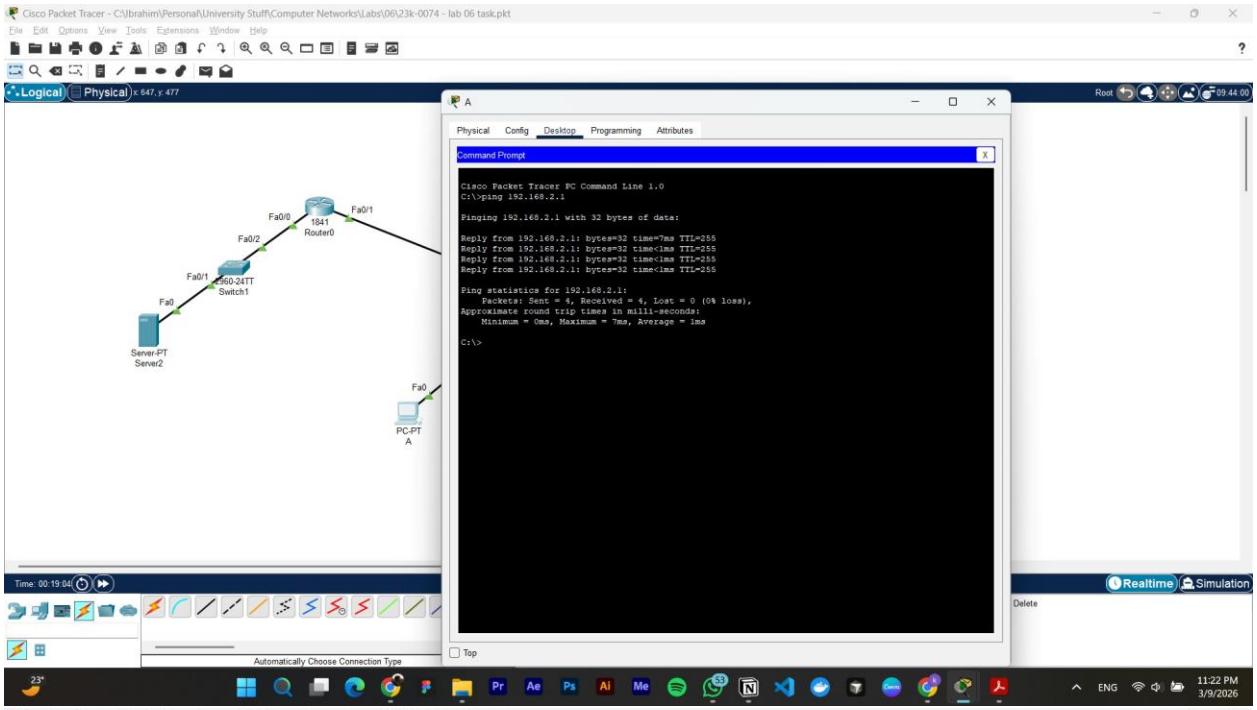

Cisco Packet Tracer
Logical
Physical
A
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address 000A.F38C.8764
IP Configuration
DHCP
Static
IPv4 Address 192.168.2.10
Subnet Mask 255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address: FE80::20A:F3FF:FE8C:8764
1841 Router0
2960-24TT Switch1
Server-PT Server2
PC-PT A
Automatically Choose Connection Type

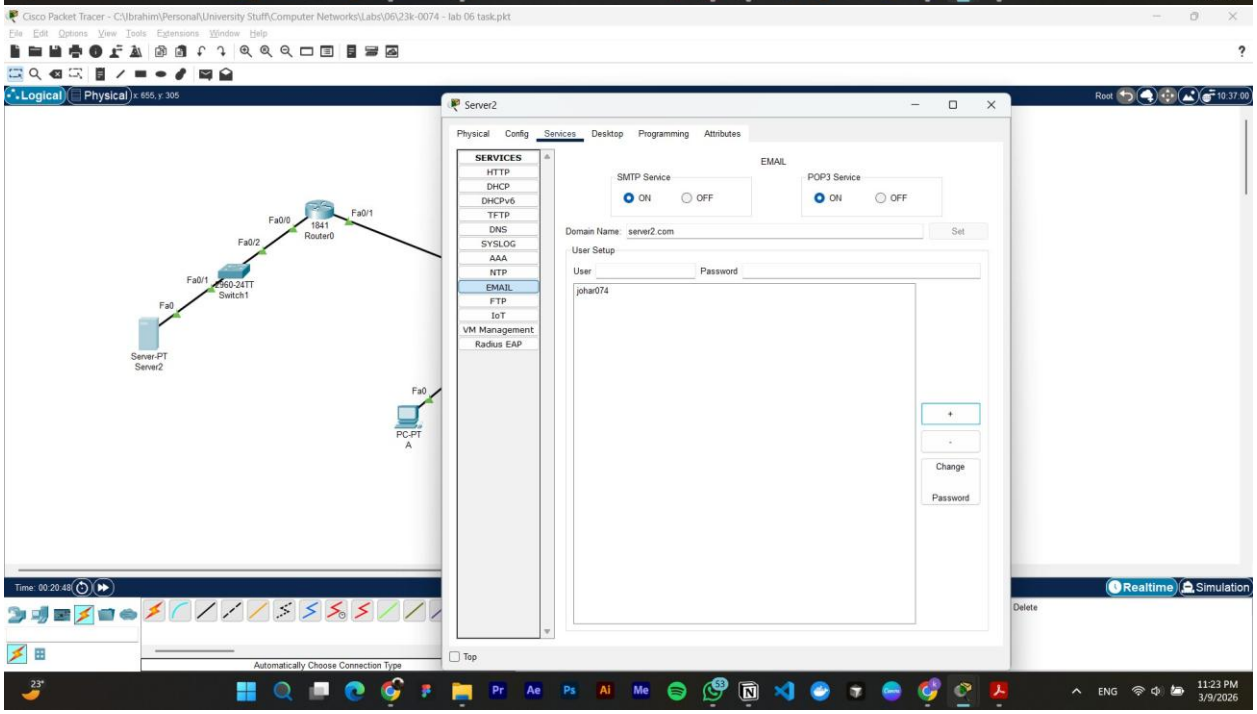

Cisco Packet Tracer
Logical
Physical
A
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name A
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 192.168.2.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Router0
Switch1
Server-PT
Server2
PC-PT
A
Simulation
Automatically Choose Connection Type

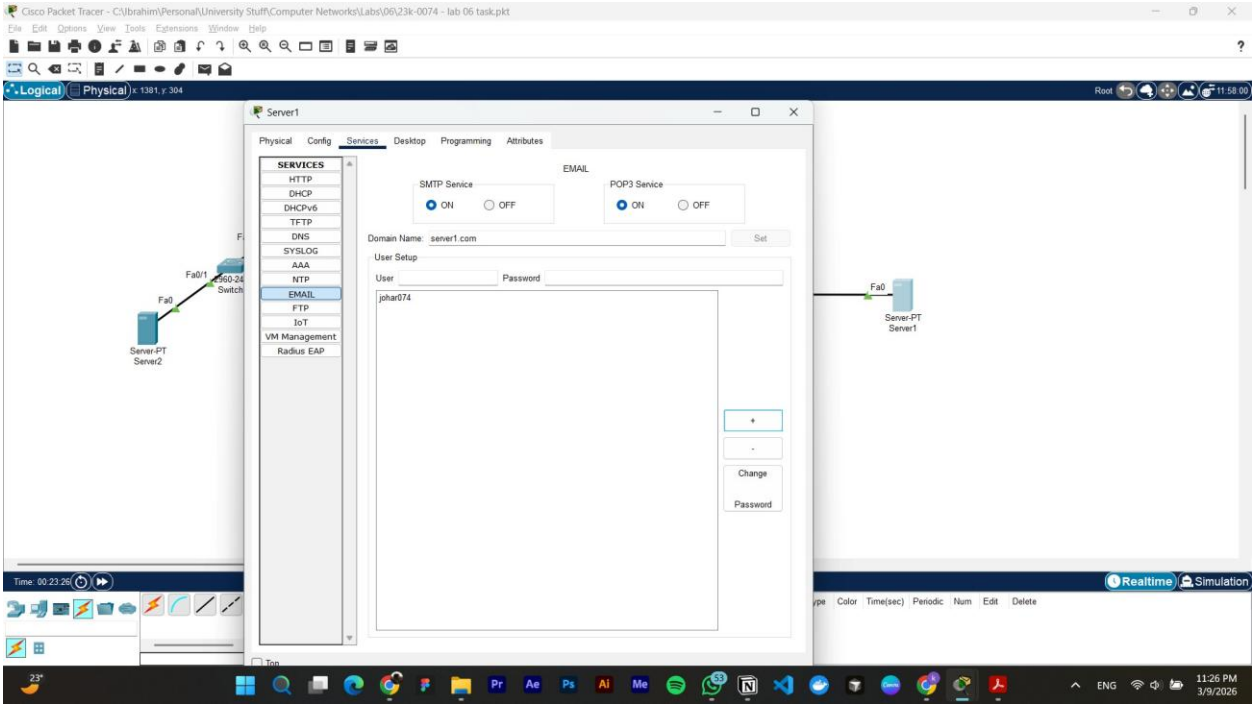

Cisco Packet Tracer
Logical
Physical
B
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address 0003:E4C9:C62E
IP Configuration
DHCP
Static
IPv4 Address 192.168.2.11
Subnet Mask 255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address: FE80::203:E4FF:FEC9:C62E
Server-PT
Server1
PC-PT
B
Realtime
Simulation

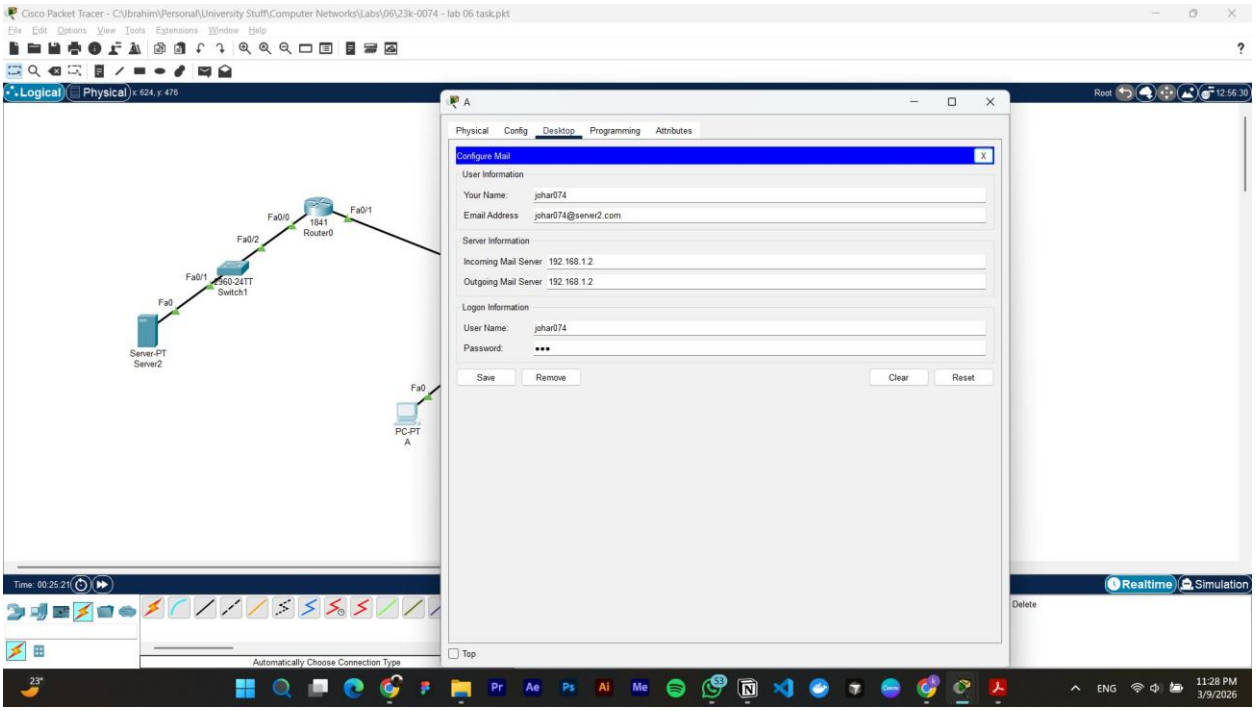

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 916, y: 246
Root 07:12:30
B
Physical Config Desktop Programming Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name B
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 192.168.2.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Fa0
Server-PT
Server1
Fa0
PC-PT
B
Realtime Simulation
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Delete
ENG
11:17 PM
3/9/2026

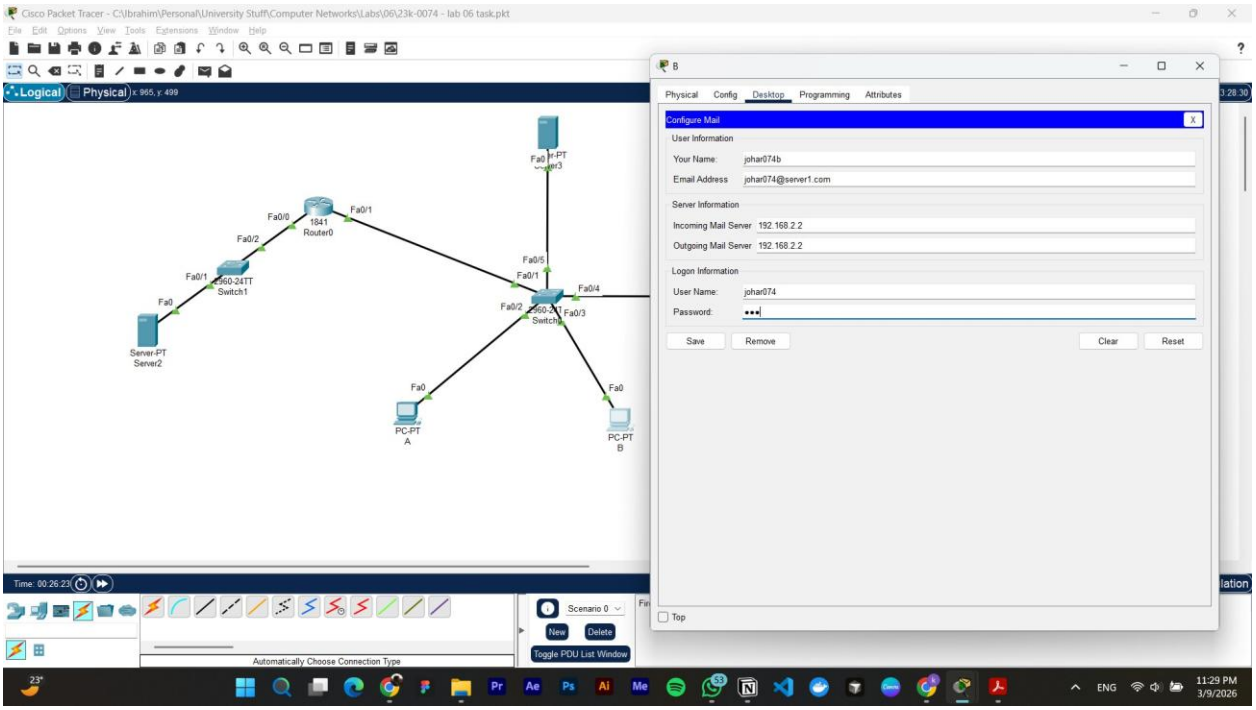

Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 636, y: 204
Root 08:57:00
Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#
Router(config-if)#
Router(config-if)#en
% Ambiguous command: "en"
Router(config)#enable
% Incomplete command.
Router(config)#config t
%Invalid hex value
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#int fa0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
Router(config)#
Copy
Paste
Top
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0
PC-PT
A
Time: 00:17:32
Automatically Choose Connection Type
New
Delete
Toggle PDU List Window
Realtime Simulation
ENG
11:20 PM
3/9/2026

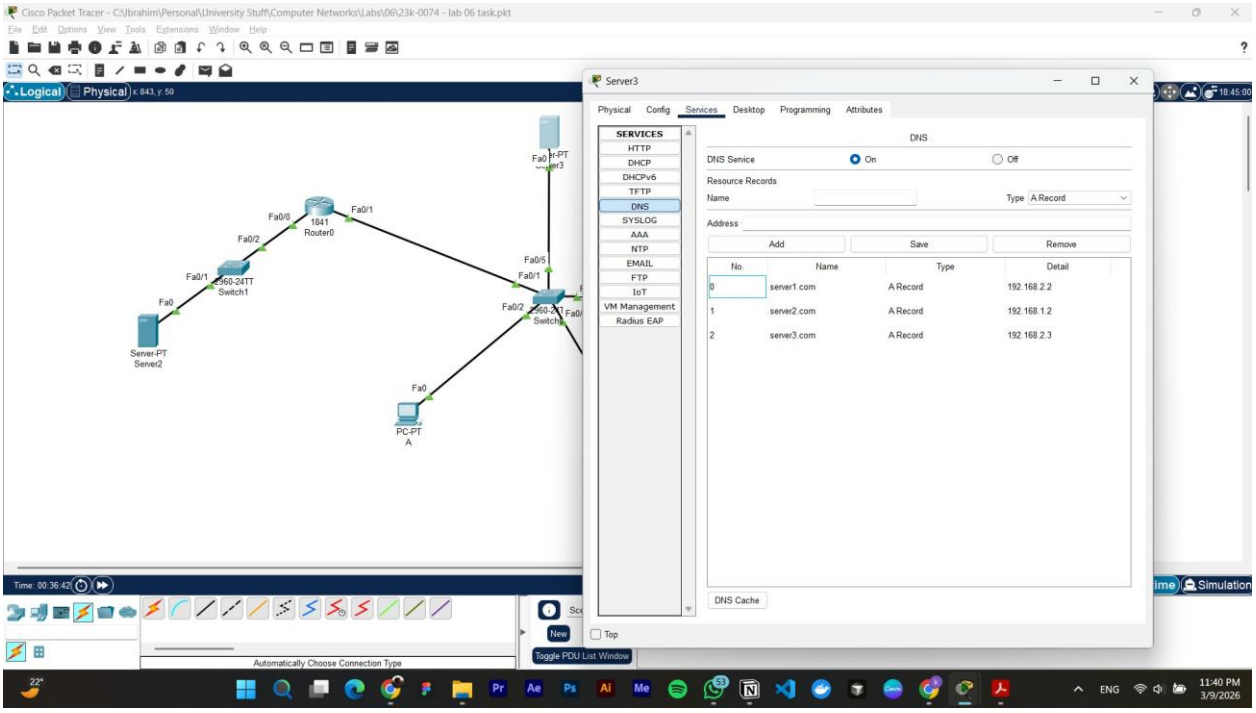

Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
Logical
Physical
A
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1
Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.2.1: bytes=32 time=7ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 7ms, Average = 1ms
C:\>
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0
PC-PT
A
Realtime
Simulation
Delete
Automatically Choose Connection Type
Top

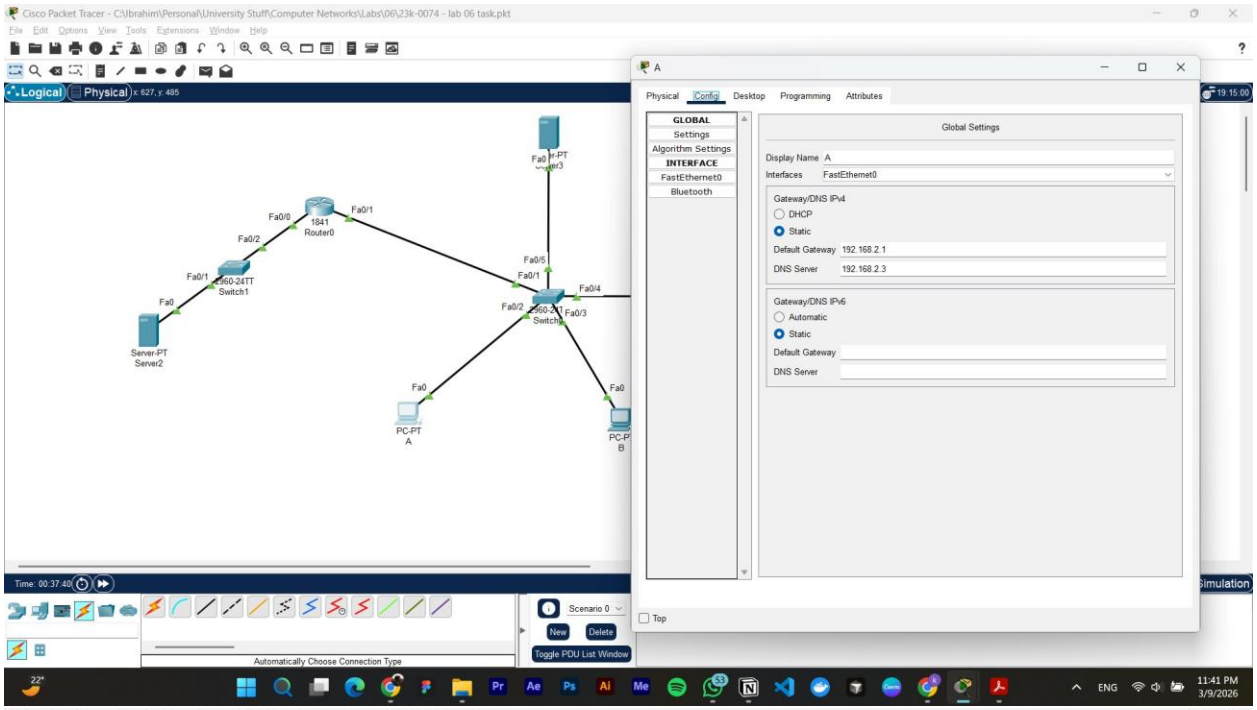

Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
Logical
Physical
Server2
Physical Config Services Desktop Programming Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
EMAIL
SMTP Service
ON
OFF
POP3 Service
ON
OFF
Domain Name: server2.com
Set
User Setup
User
Password
johar074
+
-
Change
Password
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0
PC-PT
A
Realtime
Simulation
Delete
Automatically Choose Connection Type
Top

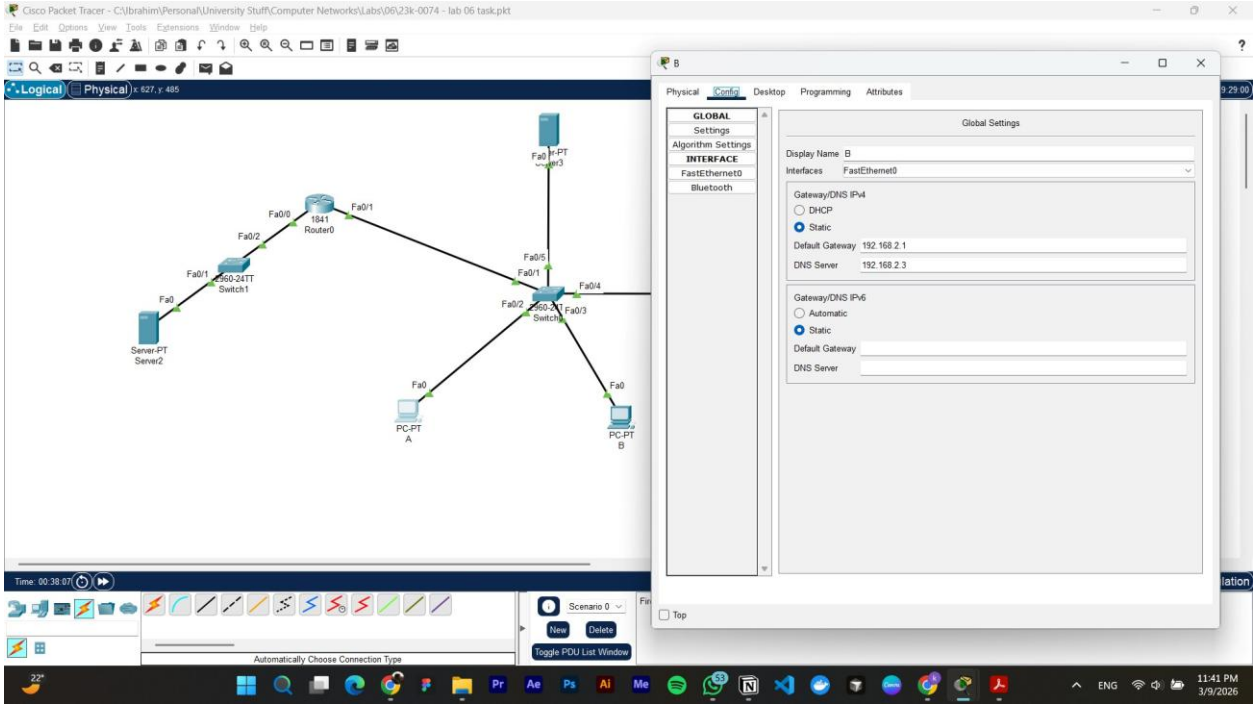

Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
Logical
Physical
Server1
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
EMAIL
SMTP Service
ON
OFF
POP3 Service
ON
OFF
Domain Name: server1.com
Set
User Setup
User
Password
johar074
Change
Password
Fa0/1
Fa0
Server-PT
Server2
Fa0
Server-PT
Server1
Realtime
Simulation

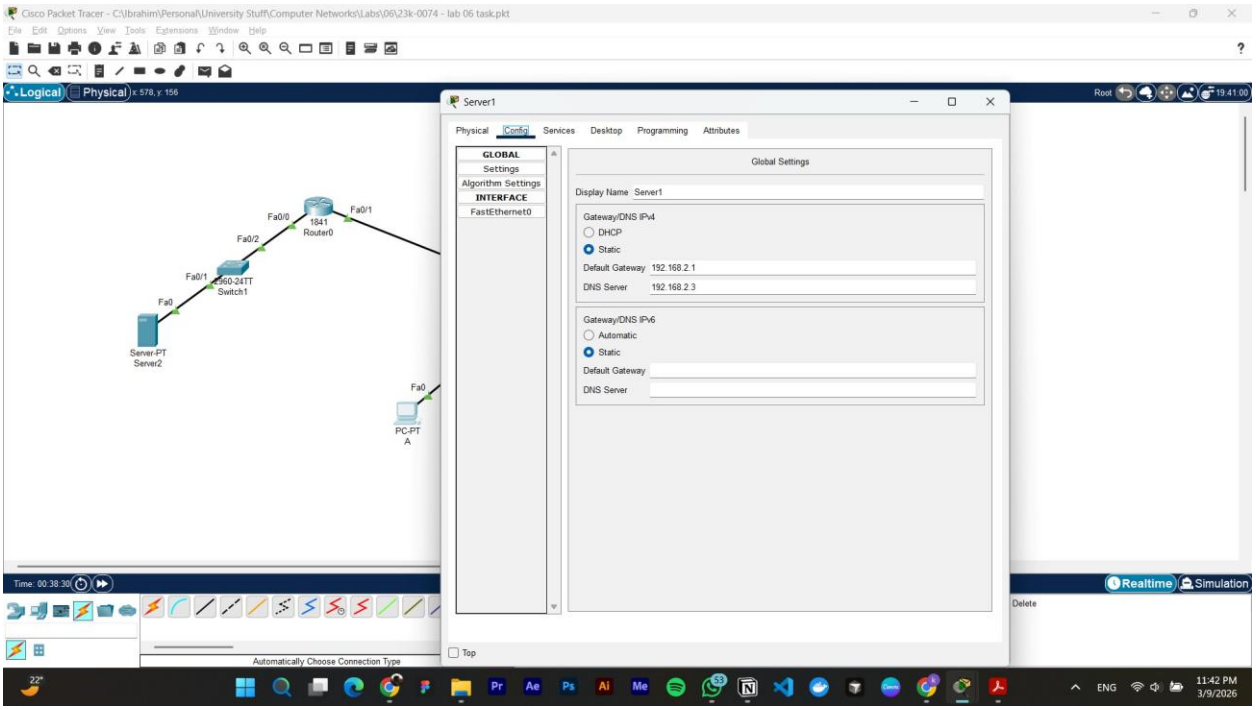

Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
Logical
Physical
A
Physical
Config
Desktop
Programming
Attributes
Configure Mail
User Information
Your Name: johar074
Email Address johar074@server2.com
Server Information
Incoming Mail Server 192.168.1.2
Outgoing Mail Server 192.168.1.2
Logon Information
User Name: johar074
Password: •••
Save
Remove
Clear
Reset
Top
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0
PC-PT
A
Automatically Choose Connection Type
Realtime
Simulation

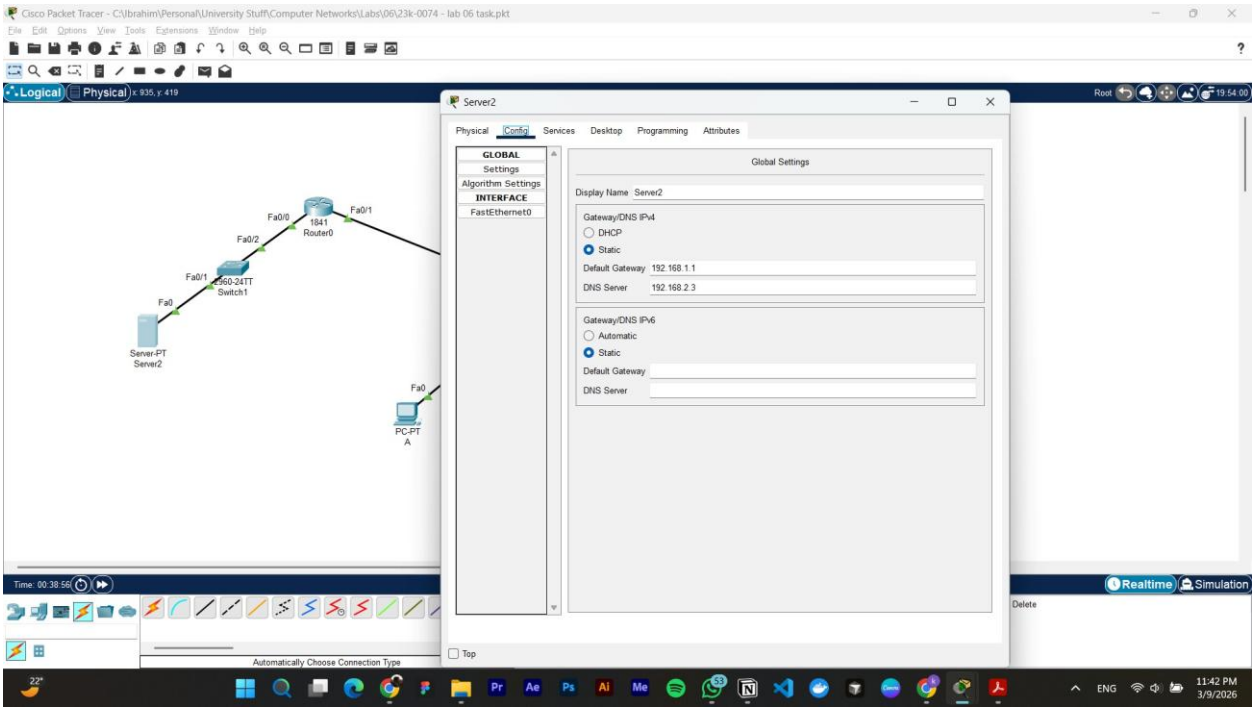

Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 965, y: 499
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0/5
Fa0/1
Fa0/4
Fa0/2
Fa0/3
Fa0
PC-PT
A
Fa0
PC-PT
B
Time: 00:26:23
Scenario 0
New
Delete
Toggle PDU List Window
Automatically Choose Connection Type
B
Physical Config Desktop Programming Attributes
Configure Mail
User Information
Your Name: johar074b
Email Address johar074@server1.com
Server Information
Incoming Mail Server 192.168.2.2
Outgoing Mail Server 192.168.2.2
Logon Information
User Name: johar074
Password: •••
Save Remove Clear Reset
Top
ENG
11:29 PM
3/9/2026

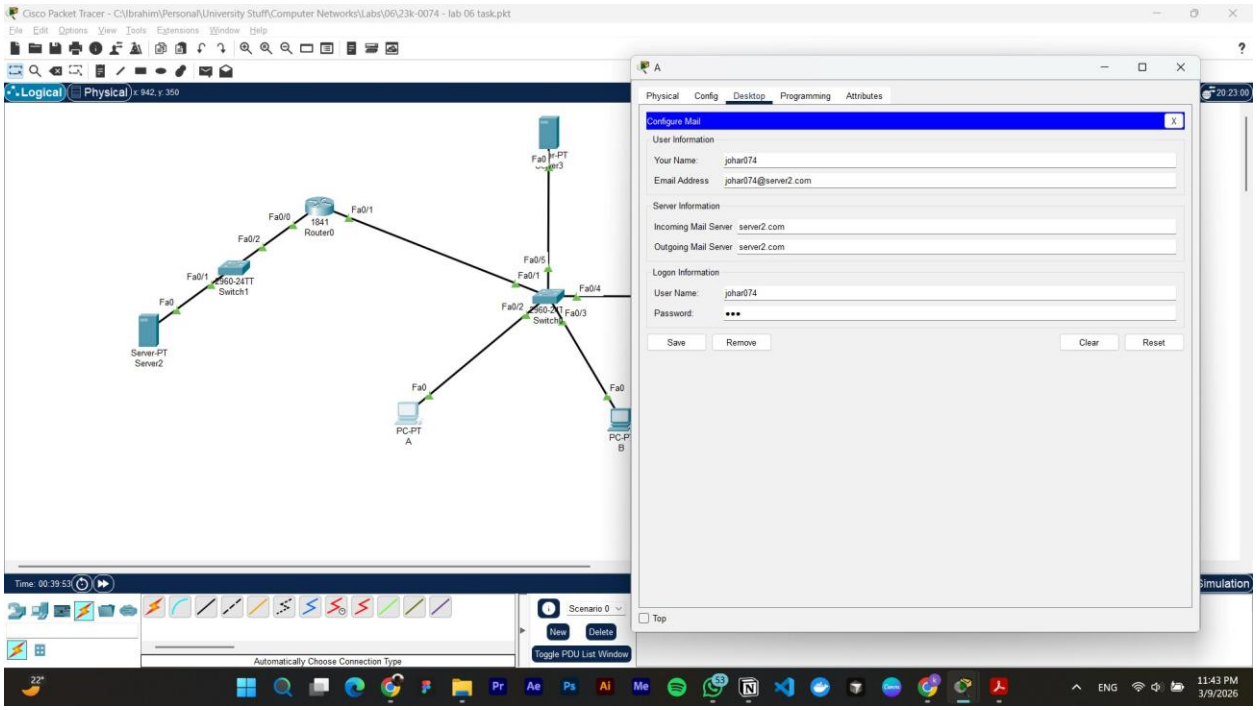

Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 843, y: 50
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0/5
Fa0/1
Fa0/2
Fa0
PC-PT
A
Time: 00:36:42
New
Toggle PDU List Window
Automatically Choose Connection Type
Server3
Physical Config Services Desktop Programming Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DNS
DNS Service On Off
Resource Records
Name
Type A Record
Address
Add Save Remove
No. Name Type Detail
0 server1.com A Record 192.168.2.2
1 server2.com A Record 192.168.1.2
2 server3.com A Record 192.168.2.3
DNS Cache
Top
ENG
11:40 PM
3/9/2026

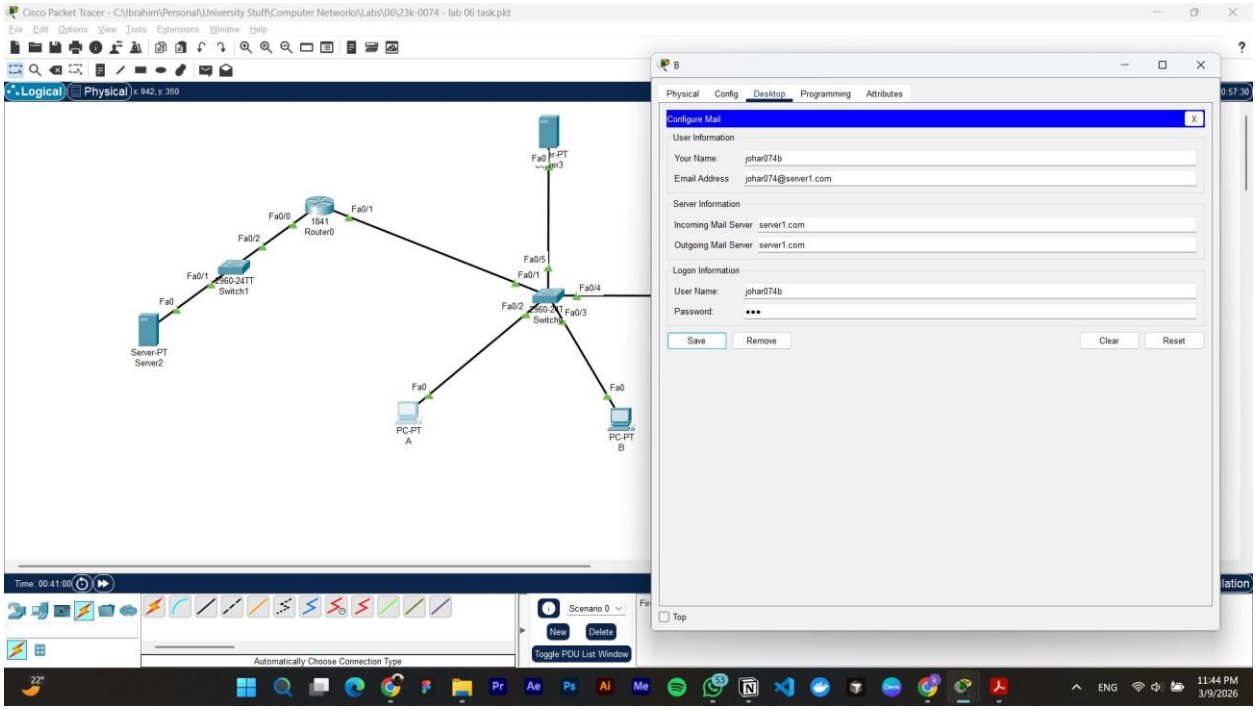

Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 627, y: 485
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0/5
Fa0/1
Fa0/4
Fa0/2
Fa0/3
Fa0
PC-PT
A
Fa0
PC-PT
B
A
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name A
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 192.168.2.1
DNS Server 192.168.2.3
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top
Time: 00:37:40
Scenario 0
New
Delete
Toggle PDU List Window
Automatically Choose Connection Type
ENG
11:41 PM
3/9/2026

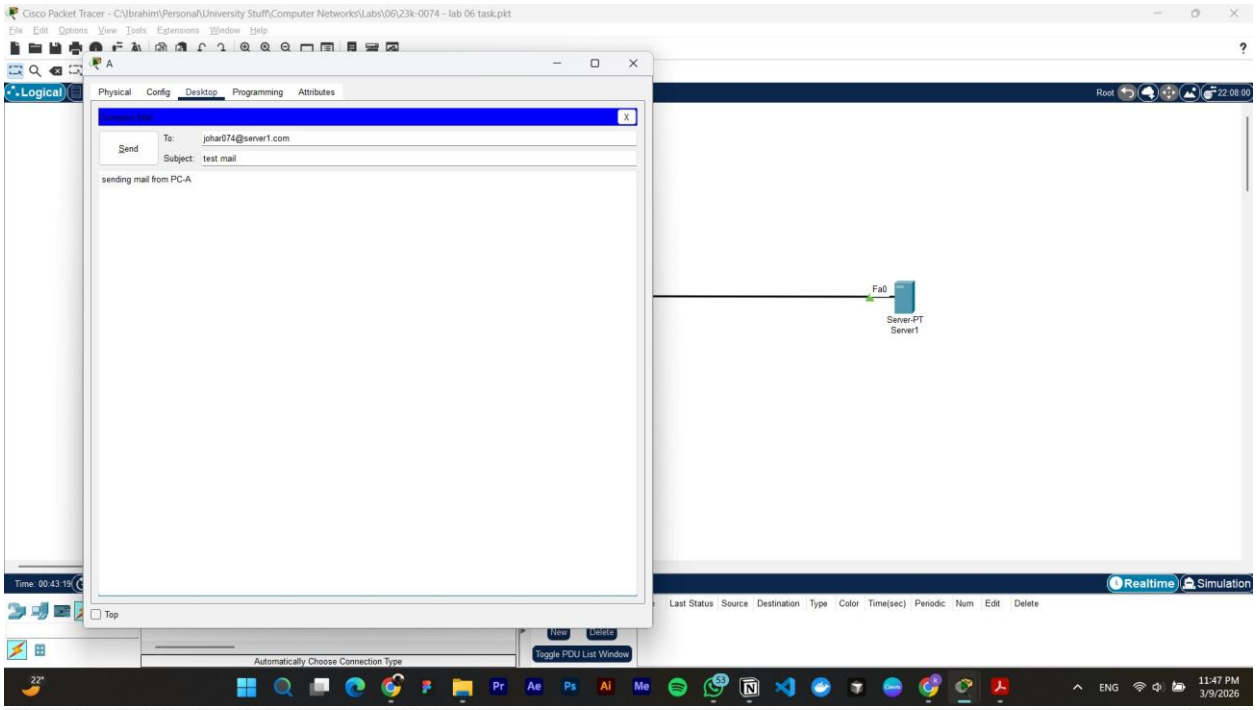

Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 627, y: 485
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0/5
Fa0/1
Fa0/4
Fa0/2
Fa0/3
Fa0
PC-PT
A
Fa0
PC-PT
B
B
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name B
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 192.168.2.1
DNS Server 192.168.2.3
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top
Time: 00:38:07
Scenario 0
New
Delete
Toggle PDU List Window
Automatically Choose Connection Type
ENG
11:41 PM
3/9/2026

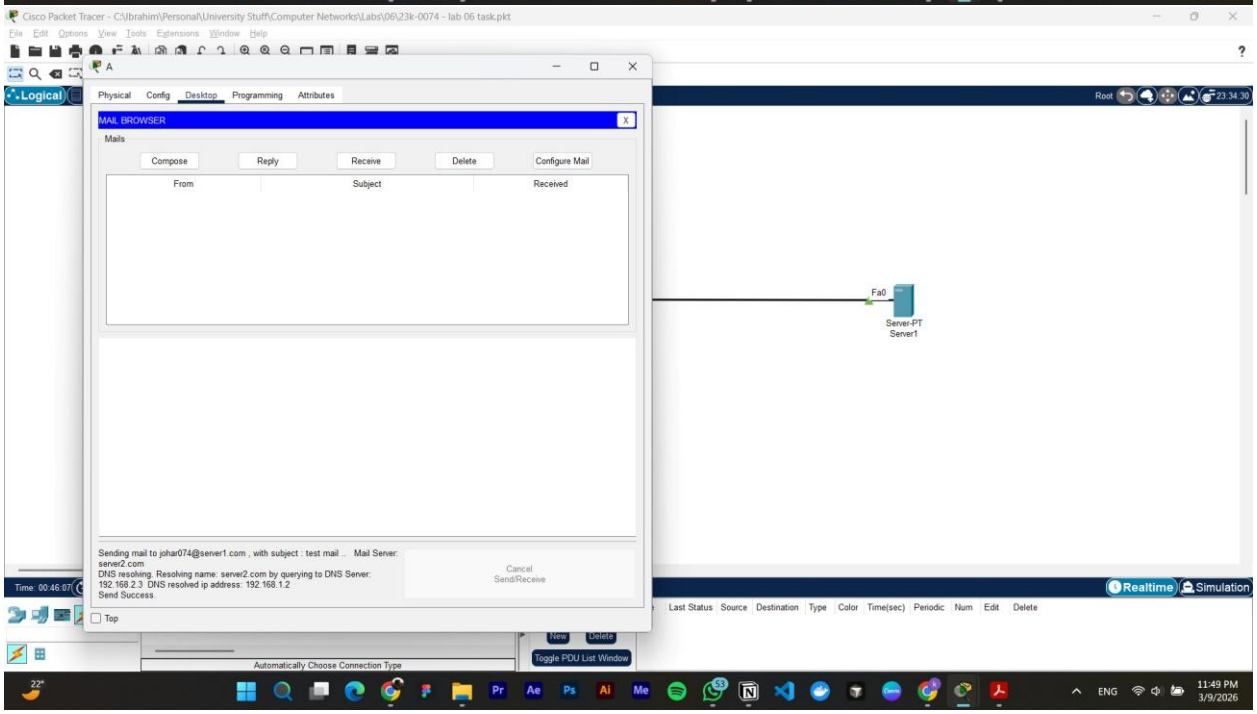

Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
Server1
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Global Settings
Display Name Server1
Gateway/DNS IPv4
DHCP
Static
Default Gateway 192.168.2.1
DNS Server 192.168.2.3
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0
PC-PT
A
Realtime
Simulation
Delete
Automatically Choose Connection Type

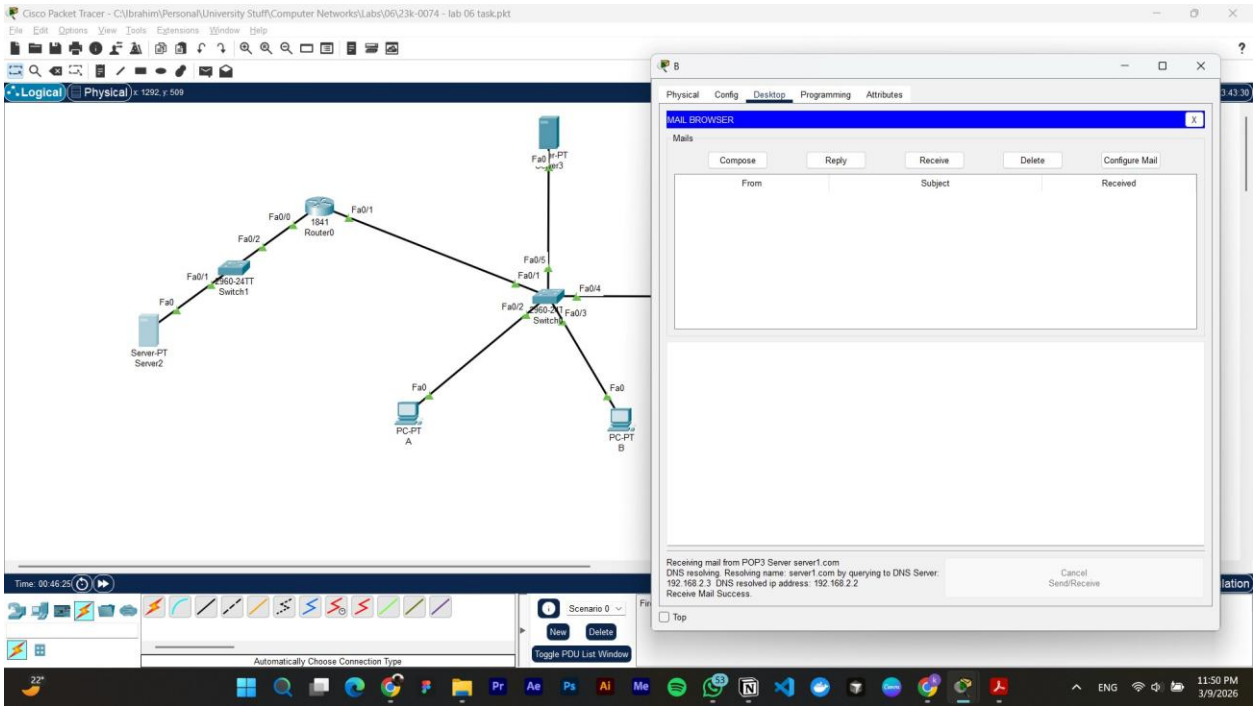

Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
Server2
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Global Settings
Display Name Server2
Gateway/DNS IPv4
DHCP
Static
Default Gateway 192.168.1.1
DNS Server 192.168.2.3
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top
Fa0/0
Fa0/1
1841
Router0
Fa0/2
Fa0/1
2960-24TT
Switch1
Fa0
Server-PT
Server2
Fa0
PC-PT
A
Realtime
Simulation
Delete
Automatically Choose Connection Type


Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 942, y: 350
Server-PT Server3
Fa0/0
Fa0/1
1841 Router0
Fa0/2
Fa0/1
2960-24TT Switch1
Fa0
Server-PT Server2
Fa0/5
Fa0/1
Fa0/4
Fa0/2
2960-24TT Switch0
Fa0/3
Fa0
PC-PT A
Fa0
PC-PT B
A
Physical Config Desktop Programming Attributes
Configure Mail
User Information
Your Name: johar074
Email Address johar074@server2.com
Server Information
Incoming Mail Server server2.com
Outgoing Mail Server server2.com
Logon Information
User Name: johar074
Password: •••
Save Remove Clear Reset
Top
Time: 00:39:53
Scenario 0
New Delete
Toggle PDU List Window
Automatically Choose Connection Type
ENG
11:43 PM
3/9/2026


Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 942, y: 350
Server-PT Server3
Fa0/0
Fa0/1
1841 Router0
Fa0/2
Fa0/1
2960-24TT Switch1
Fa0
Server-PT Server2
Fa0/5
Fa0/1
Fa0/4
Fa0/2
2960-24TT Switch0
Fa0/3
Fa0
PC-PT A
Fa0
PC-PT B
B
Physical Config Desktop Programming Attributes
Configure Mail
User Information
Your Name: johar074b
Email Address johar074@server1.com
Server Information
Incoming Mail Server server1.com
Outgoing Mail Server server1.com
Logon Information
User Name: johar074b
Password: •••
Save Remove Clear Reset
Top
Time: 00:41:00
Scenario 0
New Delete
Toggle PDU List Window
Automatically Choose Connection Type
ENG
11:44 PM
3/9/2026


Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
File Edit Options View Tools Extensions Window Help
A
Physical Config Desktop Programming Attributes
Send
To: johar074@server1.com
Subject: test mail
sending mail from PC-A
Top
Fa0
Server-PT
Server1
Logical
Root
Realtime
Simulation
Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Automatically Choose Connection Type
Toggle PDU List Window
ENG
11:47 PM
3/9/2026


Cisco Packet Tracer - C:\Ibrahim\Personal\University Stuff\Computer Networks\Labs\06\23k-0074 - lab 06 task.pkt
File Edit Options View Tools Extensions Window Help
A
Physical Config Desktop Programming Attributes
MAIL BROWSER
Mails
Compose
Reply
Receive
Delete
Configure Mail
From
Subject
Received
Sending mail to johar074@server1.com , with subject : test mail ... Mail Server: server2.com
DNS resolving. Resolving name: server2.com by querying to DNS Server: 192.168.2.3 DNS resolved ip address: 192.168.1.2
Send Success.
Cancel Send/Receive
Top
Fa0
Server-PT
Server1
Logical
Root
Realtime
Simulation
Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Automatically Choose Connection Type
Toggle PDU List Window
ENG
11:49 PM
3/9/2026

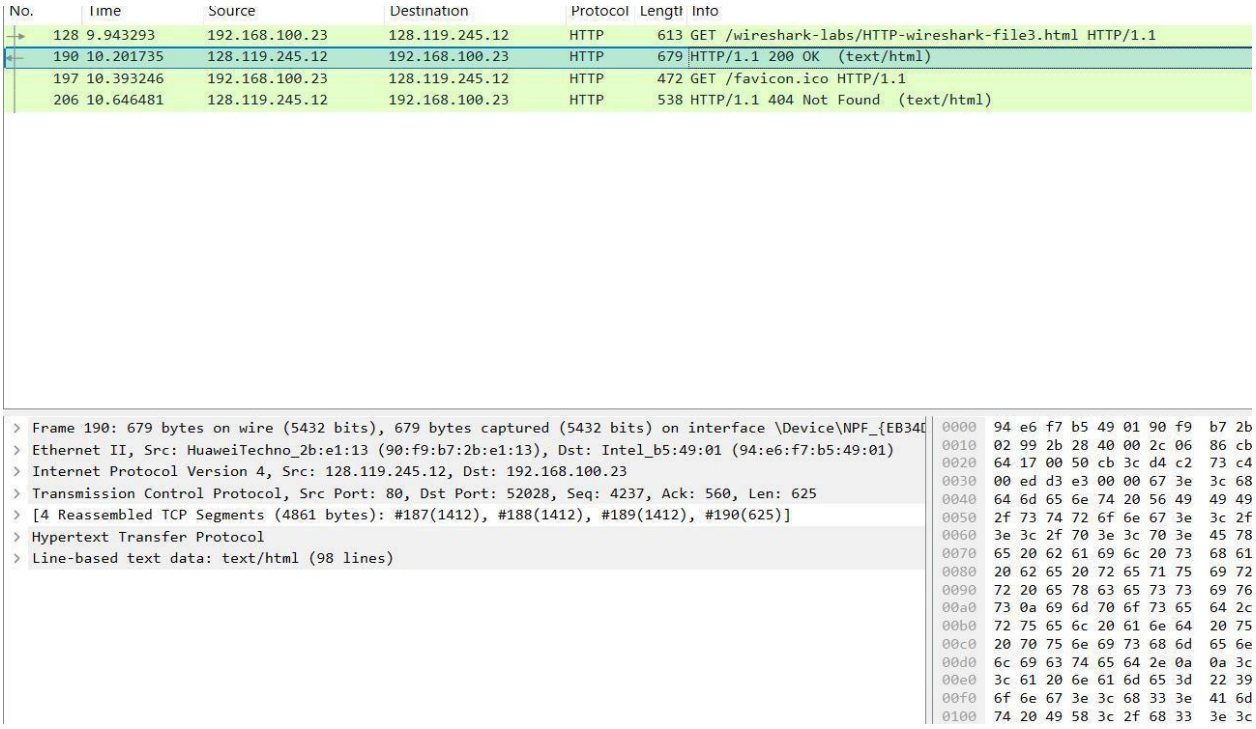


3) Retrieving Very Long Docs

128	9.943293	192.168.100.23	128.119.245.12	HTTP	613	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
190	10.201735	128.119.245.12	192.168.100.23	HTTP	679	HTTP/1.1 200 OK (text/html)
197	10.393246	192.168.100.23	128.119.245.12	HTTP	472	GET /favicon.ico HTTP/1.1
206	10.646481	128.119.245.12	192.168.100.23	HTTP	538	HTTP/1.1 404 Not Found (text/html)

```
> Frame 128: 613 bytes on wire (4904 bits), 613 bytes captured (4904 bits) on interface \Device\NPF_{EB34D
> Ethernet II, Src: Intel_b5:49:01 (94:e6:f7:b5:49:01), Dst: HuaweiTechno_2b:e1:13 (90:f9:b7:2b:e1:13)
> Internet Protocol Version 4, Src: 192.168.100.23, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 52028, Dst Port: 80, Seq: 1, Ack: 1, Len: 559
> Hypertext Transfer Protocol
```

```
0000  90 f9 b7 2b e1 13 94 e6  f7 b5 49 01 08
0010  02 57 73 a4 40 00 80 06  00 00 c0 a8 64
0020  f5 0c cb 3c 00 50 52 bf  02 dd d4 c2 63
0030  00 ff 9c 8d 00 00 47 45  54 20 2f 77 69
0040  68 61 72 6b 2d 6c 61 62  73 2f 48 54 54
0050  69 72 65 73 68 61 72 6b  2d 66 69 6c 65
0060  74 6d 6c 20 48 54 54 50  2f 31 2e 31 0d
0070  73 74 3a 20 67 61 69 61  2e 63 73 2e 75
0080  73 2e 65 64 75 0d 0a 43  6f 6e 6e 65 63
0090  6e 3a 20 6b 65 65 70 2d  61 6c 69 76 65
00a0  70 67 72 61 64 65 2d 49  6e 73 65 63 75
00b0  52 65 71 75 65 73 74 73  3a 20 31 0d 0a
00c0  72 2d 41 67 65 6e 74 3a  20 4d 6f 7a 69
00d0  2f 35 2e 30 20 28 57 69  6e 64 6f 77 73
00e0  20 31 30 2e 30 3b 20 57  69 6e 36 34 3b
00f0  34 29 20 41 70 70 6c 65  57 65 62 4b 69
0100  33 37 2e 33 36 20 28 4b  48 54 4d 4c 2c
```



1. How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message forthe Bill or Rights?

The browser sent 2 GET requests; packet 128 and packet 197. The GET message for bill of rights is in packet 128.

2. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

Packet 190

3. What is the status code and phrase in the response?

200 OK

4. How many data-containing TCP segments were needed to carry the single HTTP response and the text ofthe Bill of Rights?

4 segments ; 187,188,189,190

The packets were assembled into the final packet 190 which is the response to the GET bill of rights

4) HTML docs with embedded objects

1179	50.799066	192.168.100.23	128.119.245.12	HTTP	526	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
1185	50.997270	128.119.245.12	192.168.100.23	HTTP	1355	HTTP/1.1 200 OK (text/html)
1189	51.056282	192.168.100.23	128.119.245.12	HTTP	472	GET /pearson.png HTTP/1.1
1204	51.250220	128.119.245.12	192.168.100.23	HTTP	841	HTTP/1.1 200 OK (PNG)
1208	51.280517	192.168.100.23	178.79.137.164	HTTP	439	GET /8E_cover_small.jpg HTTP/1.1
1214	51.462971	178.79.137.164	192.168.100.23	HTTP	225	HTTP/1.1 301 Moved Permanently

```
> Frame 1179: 526 bytes on wire (4208 bits), 526 bytes captured (4208 bits) on interface \Device\NPF_{EB34
> Ethernet II, Src: Intel_b5:49:01 (94:e6:f7:b5:49:01), Dst: HuaweiTechno_2b:e1:13 (90:f9:b7:2b:e1:13)
> Internet Protocol Version 4, Src: 192.168.100.23, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 58670, Dst Port: 80, Seq: 1, Ack: 1, Len: 472
> Hypertext Transfer Protocol
```

```
0000  90 f9 b7 2b e1 13 94 e6  f7 b5 49
0010  02 00 74 3a 40 00 80 06  00 00 c0
0020  f5 0c e5 2e 00 50 01 24  57 88 73
0030  00 ff 9c 36 00 00 47 45  54 20 2f
0040  68 61 72 6b 2d 6c 61 62  73 2f 48
0050  69 72 65 73 68 61 72 6b  2d 66 69
0060  74 6d 6c 20 48 54 54 50  2f 31 2e
0070  73 74 3a 20 67 61 69 61  2e 63 73
0080  73 2e 65 64 75 0d 0a 43  6f 6e 6e
0090  6e 3a 20 6b 65 65 70 2d  61 6c 69
00a0  70 67 72 61 64 65 2d 49  6e 73 65
00b0  52 65 71 75 65 73 74 73  3a 20 31
00c0  72 2d 41 67 65 6e 74 3a  20 4d 6f
00d0  2f 35 2e 30 20 28 57 69  6e 64 6f
00e0  20 31 30 2e 30 3b 20 57  69 6e 36
00f0  34 29 20 41 70 70 6c 65  57 65 62
0100  33 37 2e 33 36 20 28 4b  48 54 4d
```

1. How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?

3 GET requests are sent.
2 requests were sent to 128.119.245.12 1 request was sent to 178.79.137.164

2. Can you tell whetheryour browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain.

In parallel, because the GET requests for the two images were sent at a very small time interval.

5) HTTP Authentication

No.	Time	Source	Destination	Protocol	Length	Info
599	40.027941	192.168.100.23	128.119.245.12	HTTP	541	GET /wireshark-labs/protected_pages/HTTP-wiresharkfile5.html HTTP/1.1
605	40.689208	128.119.245.12	192.168.100.23	HTTP	771	HTTP/1.1 401 Unauthorized (text/html)
678	48.441425	192.168.100.23	128.119.245.12	HTTP	626	GET /wireshark-labs/protected_pages/HTTP-wiresharkfile5.html HTTP/1.1
684	49.060116	128.119.245.12	192.168.100.23	HTTP	583	HTTP/1.1 404 Not Found (text/html)

```
> Frame 599: 541 bytes on wire (4328 bits), 541 bytes captured (4328 bits) on interface \Device\NPF_{EB34D
> Ethernet II, Src: Intel_b5:49:01 (94:e6:f7:b5:49:01), Dst: HuaweiTechno_2b:e1:13 (90:f9:b7:2b:e1:13)
> Internet Protocol Version 4, Src: 192.168.100.23, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 62826, Dst Port: 80, Seq: 1, Ack: 1, Len: 487
> Hypertext Transfer Protocol
```

```
0000  90 f9 b7 2b e1 13 94 e6  f7 b5 49 01 08 00
0010  02 0f 74 8d 40 00 80 06  00 00 c0 a8 64 17
0020  f5 0c f5 6a 00 50 0b f7  97 a6 1b 64 62 15
0030  00 ff 9c 45 00 00 47 45  54 20 2f 77 69 72
0040  68 61 72 6b 2d 6c 61 62  73 2f 70 72 6f 74
0050  74 65 64 5f 70 61 67 65  73 2f 48 54 54 50
0060  69 72 65 73 68 61 72 6b  66 69 6c 65 35 2e
0070  6d 6c 20 48 54 54 50 2f  31 2e 31 0d 0a 48
0080  74 3a 20 67 61 69 61 2e  63 73 2e 75 6d 61
0090  2e 65 64 75 0d 0a 43 6f  6e 6e 65 63 74 69
00a0  3a 20 6b 65 65 70 2d 61  6c 69 76 65 0d 0a
00b0  67 72 61 64 65 2d 49 6e  73 65 63 75 72 65
00c0  65 71 75 65 73 74 73 3a  20 31 0d 0a 55 73
```

1. What is the server's response (status code and phrase) in response to the initial HTTP GET message from your browser?

401 Unauthorized

2. When your browser's sends the HTTP GET message for the second time, what new field is included in the HTTP GET message?

Authorization header field (which contains the username and password entered by the user)